I-Logical Approaches Circuits

Foundations of Neuro-Symbolic AI

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Outline

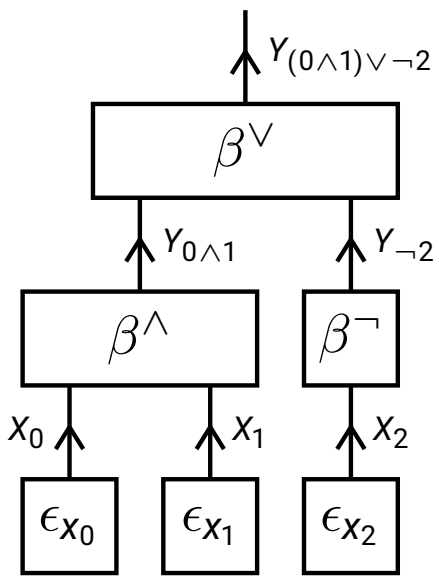
- Circuit computation model
- Tensor representation
- Example: m-adic sums

Recall: Basis calculus in propositional logic

Evaluation of propositional formulas at states in basis calculus:

- Start with the one-hot encoding state to be evaluated
- Evaluate iteratively sub-formulas, pass the result to the more complex sub-formulas

Example: The formula $(X_0 \wedge X_1) \vee \neg X_2$ is evaluated at a state $(x_0, x_1, x_2) \in [2]^3$ by



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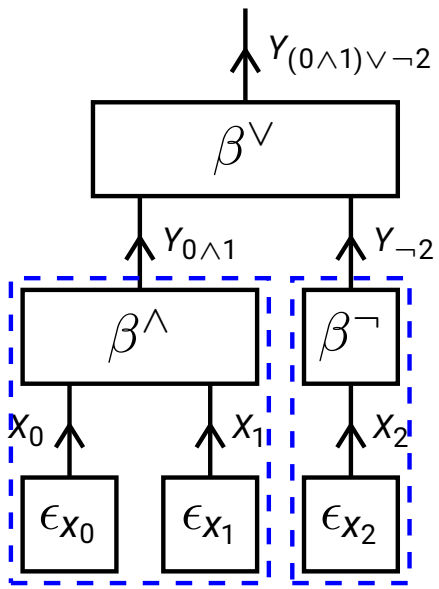

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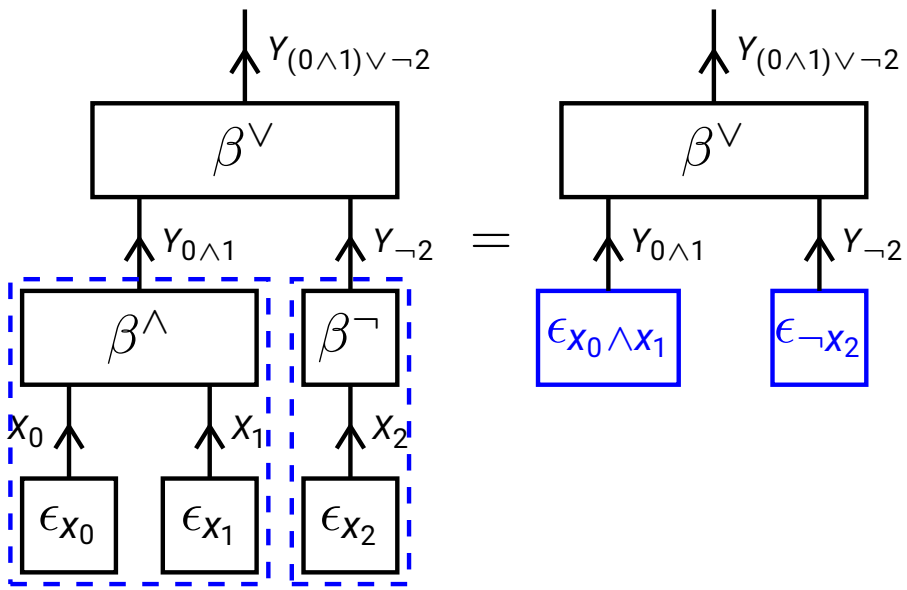

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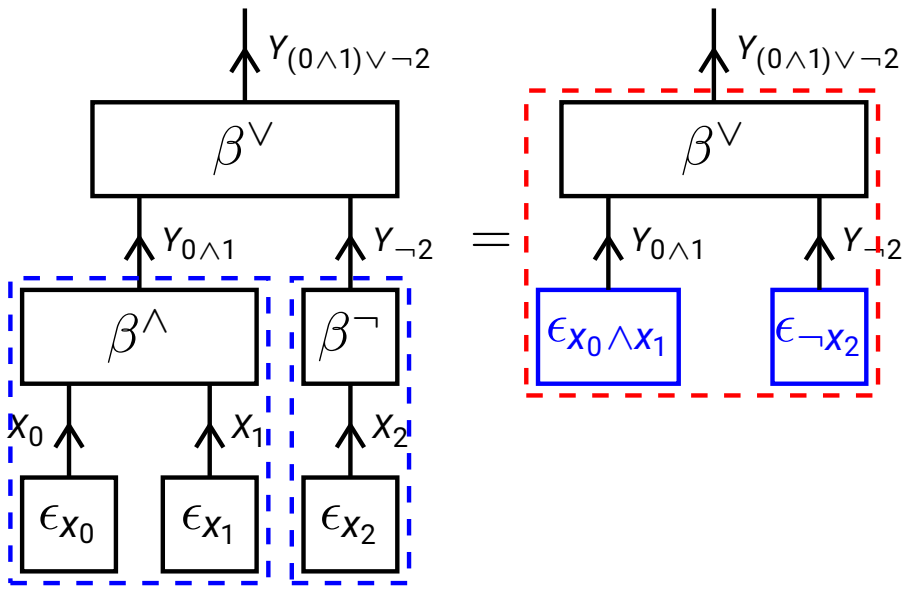

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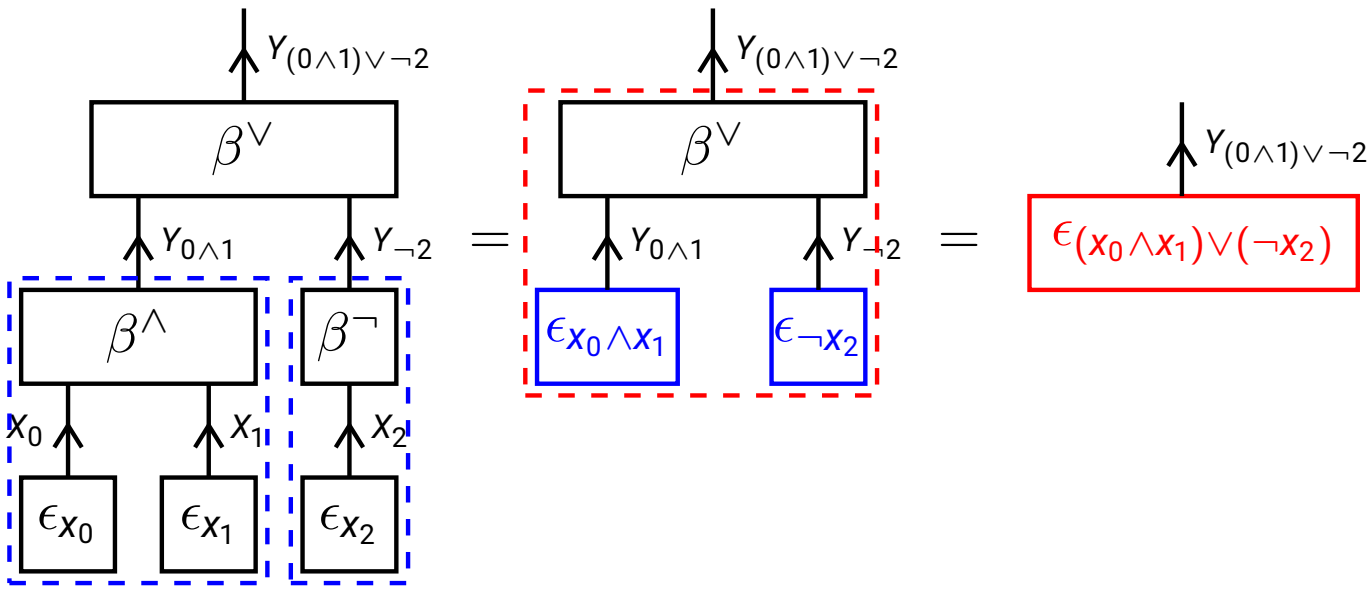

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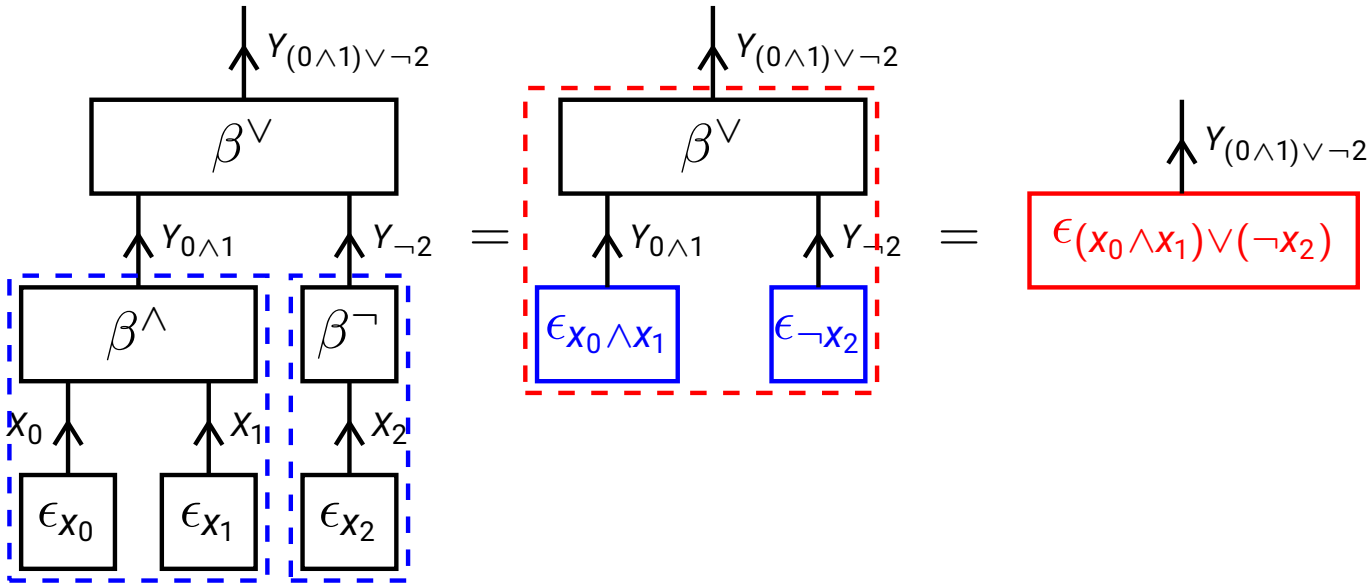


Can we generalize to arbitrary functions and find tensor network decompositions based on circuits?

Generalization to circuits

A generic tensor has arbitary leg dimensions and represents a function

$$q : x_{[d]} \to \tau \left[X_{[d]} = x_{[d]} \right]$$

How to find a decomposition of a tensor into smaller tensors, based on a syntactical description of the function represented by the tensor?

Generalizations required:

- Arbitrary state numbers to variables: Leaving the interpretation of the states [2] as $\{\mathrm{False}, \mathrm{True}\}$, use instead state sets $[m]$
- Arbitrary functions applied on states

Circuits

Definition (Circuit)

A circuit is a decorated hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where:

- Each node $v \in \mathcal{V}$ is decorated by a dimension $m_v \in \mathbb{N}$
- Each edge $e \in \mathcal{E}$ is a tuple of incoming nodes e^{in} and outgoing nodes e^{out}
- Each edge $e \in \mathcal{E}$ is decorated by a function

$$q_e : \bigtimes_{v \in e^{\text{in}}} [m_v] \to \bigtimes_{v \in e^{\text{out}}} [m_v]$$

We assume that:

- The hypergraph is acyclic
- Each node appears at most once as an outgoing node

Functions represented by circuits

We denote by

- $\mathcal{V}^{\text{in}}$ the nodes appearing never as outgoing nodes.
- $\mathcal{V}^{\text{out}}$ the nodes appearing never as incoming nodes.

Now we can iteratively define a function

$$g^v : \bigtimes_{\tilde{v} \in e^{\text{in}}} [m_{\tilde{v}}] \to [m_v]$$

as the composition of the functions decorating the edges, mapping the states of $\mathcal{V}^{\text{in}}$ to the states of $\mathcal{V}^{\text{out}}$, by

$$g^v = q_e|_v \circ \left(\bigtimes_{\tilde{v} \in e^{\text{in}}} g^{\tilde{v}} \right).$$

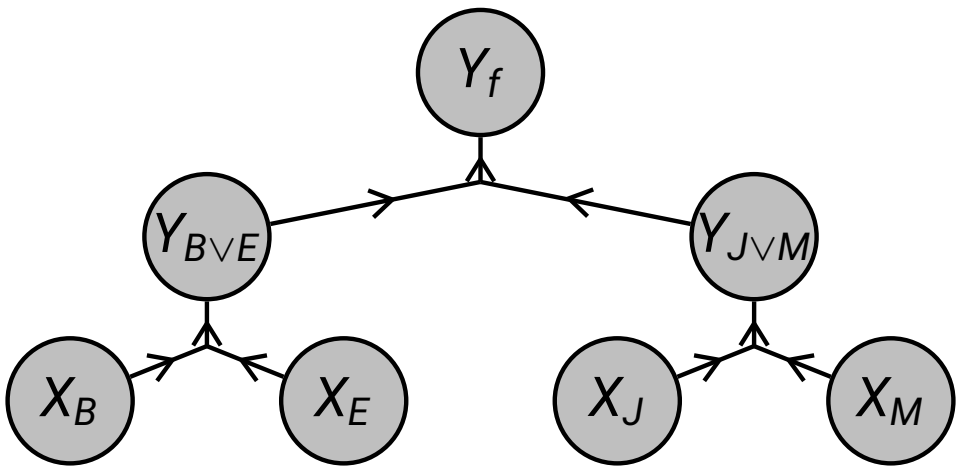
The circuit represents the function

$$g^{\mathcal{G}} = \bigtimes_{v \in \mathcal{V}^{\text{out}}} g^v .$$

Example: Propositional syntax

Consider the formula $(X_B \vee X_E) \Rightarrow (X_J \vee X_M)$

- The nodes are the variables X_B, X_E, X_J, X_M, the sub-formulas $Y_{B \vee E}, Y_{J \vee M}$, and the formula Y_f
- The nodes $\mathcal{V}^{\text{in}} = \{X_B, X_E, X_J, X_M\}$ do not appear as outgoing nodes
- The node Y_f is the only one not appearing as incoming nodes, thus $\mathcal{V}^{\text{out}} = \{Y_f\}$



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- Tensor representation
- Example: m-adic sums

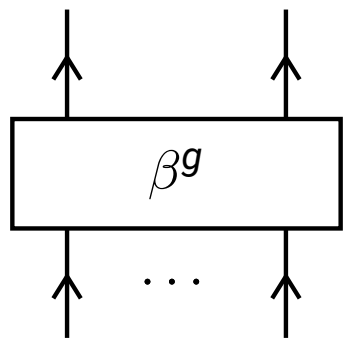
General basis encodings

Consider an arbitrary function

$$g : \bigtimes_{v \in \mathcal{V}^{\text{in}}} [m_v] \to \bigtimes_{v \in \mathcal{V}^{\text{out}}} [m_v] .$$

We define its basis encoding as the tensor

$$\beta^g [X_{\mathcal{V}^{\text{out}}}, X_{\mathcal{V}^{\text{in}}}] = \sum_{x_{\mathcal{V}^{\text{in}}} \in \bigtimes_{v \in \mathcal{V}^{\text{in}}} [m_v]} \epsilon_{g(x_{\mathcal{V}^{\text{in}}})} [X_{\mathcal{V}^{\text{out}}}] \otimes \epsilon_{x_{\mathcal{V}^{\text{in}}}} [X_{\mathcal{V}^{\text{in}}}] .$$



Decomposition into tensor networks

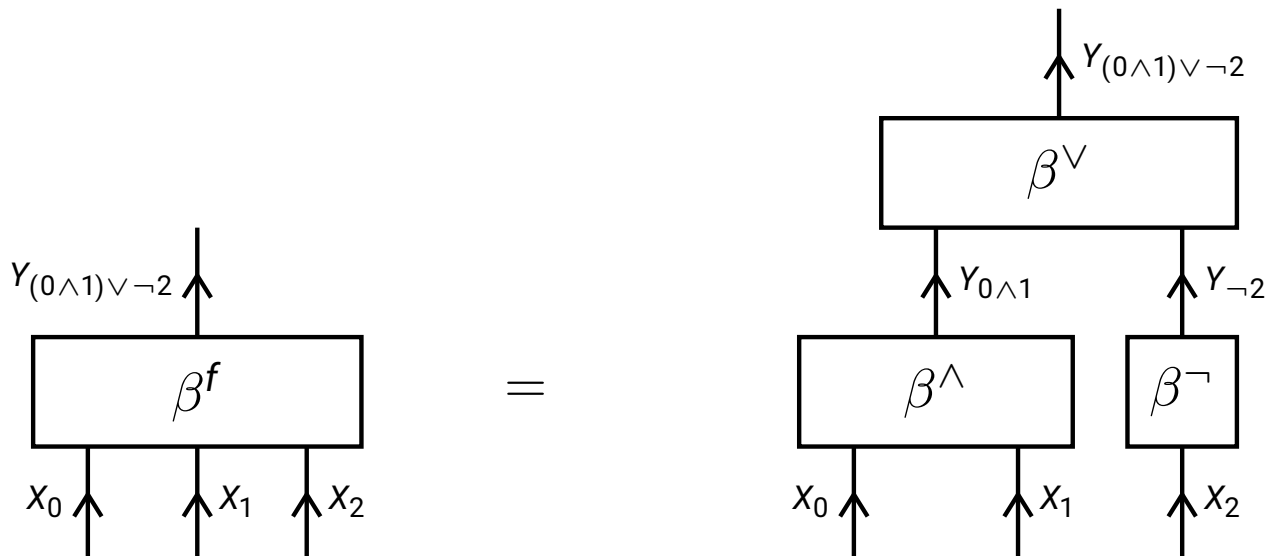
To each edge e we choose a basis encoding

$$\beta^{q_e}\left[X_{e^{\text{out}}}, X_{e^{\text{in}}}\right] = \sum_{x_{e^{\text{in}}} \in \times_{v \in e^{\text{in}}}[m_v]} \epsilon_{q_e(x_{e^{\text{in}}})}\left[X_{e^{\text{out}}}\right] \otimes \epsilon_{x_{e^{\text{in}}}}\left[X_{e^{\text{in}}}\right]$$

The basis encoding to a function of a circuit is decomposed into basis encodings of the node functions:

$$\beta^{q_{\mathcal{G}}}\left[X_{\mathcal{V}^{\text{out}}}, X_{\mathcal{V}^{\text{in}}}\right] = \left\langle \left\{ \beta^{q_e}\left[X_{e^{\text{out}}}, X_{e^{\text{in}}}\right] : e \in \mathcal{E} \right\} \right\rangle_{\left[X_{\mathcal{V}^{\text{out}}}, X_{\mathcal{V}^{\text{in}}}\right]}$$

Example: The basis encoding of the formula $f = (X_0 \wedge X_1) \vee \neg X_2$ is decomposed into the basis encodings of the connectives $\wedge, \vee, \neg$ as



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m-adic representation of integers

Let m, d be arbitrary natural numbers. Any number $n \in [m^d]$ has a unique m-adic representation

$$\sum_{k \in [d]} x_k \cdot m^k$$

where for each $k \in [d]$

$$x_k \in [m] \,.$$

Example: For $m = 3$ and $d = 2$, the number $5 \in [9]$ has the 3-adic representation

$$(x_0, x_1) = (2, 1)$$

since

$$5 = 2 \cdot 3^0 + 1 \cdot 3^1 \,.$$

Summation function

We want to find a circuit decomposing the summation function

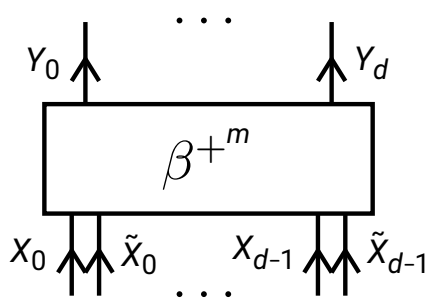
$$+^m \ : \ [m^d] \times [m^d] \to [m^{d+1}]$$

defined for tuples $(x_{[d]}, \tilde{x}_{[d]}) \in [m]^d$ with $+^m(x_{[d]}, \tilde{x}_{[d]}) = y_{[d+1]}$ as

$$\sum_{k \in [d+1]} y_k \cdot m^k = \sum_{k \in [d]} (x_k + \tilde{x}_k) \cdot m^k \ .$$

The basis encoding of the sum is the tensor

$$\beta^{+^m} \left[Y_{[d+1]}, X_{[d]}, \tilde{X}_{[d]} \right] = \sum_{x_{[d]}, \tilde{x}_{[d]}} \epsilon_{I^{-1}(I(x_{[d]}) + I(\tilde{x}_{[d]}))} \left[Y_{[d+1]} \right] \otimes \epsilon_{x_{[d]}} \left[X_{[d]} \right] \otimes \epsilon_{\tilde{x}_{[d]}} \left[\tilde{X}_{[d]} \right] \ .$$



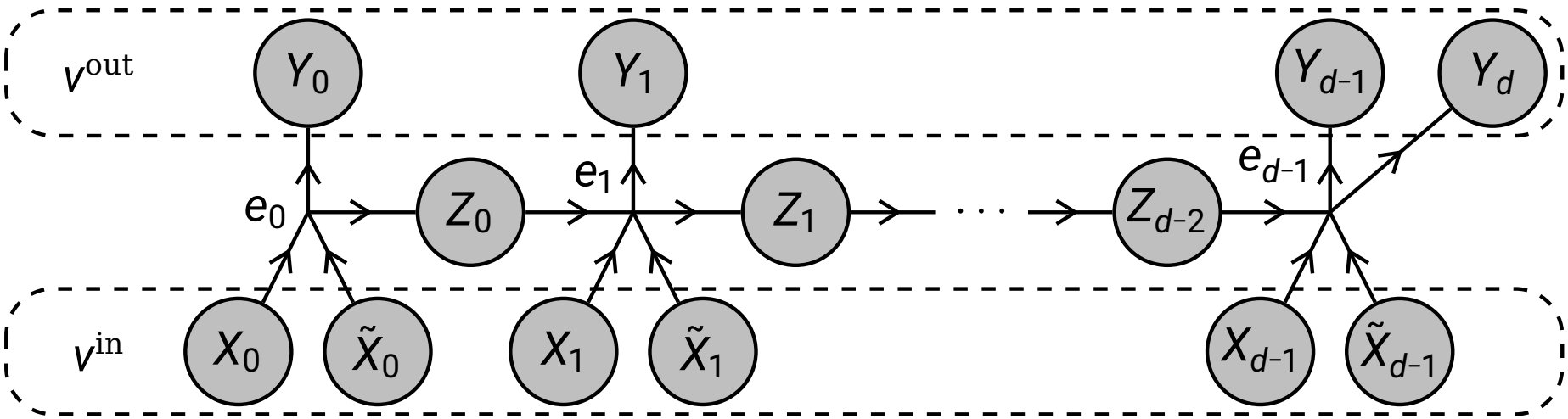
Circuit of half-adders

Nodes representing the bits to numbers:

- $X_{[d]}$ representing the first number to be summed
- $\tilde{X}_{[d]}$ representing the second number to be summed
- $Y_{[d+1]}$ representing the sum
- $Z_{[d-1]}$ representing carry bits

Edges are enumerated by $[d-1] = \{0, \ldots, d-2\}$ representing half-adders:

$$q_{e_k}(z_{k-1}, x_k, \tilde{x}_k) = \left((z_{k-1} + x_k + \tilde{x}_k) \mod m, \left\lfloor \frac{z_{k-1} + x_k + \tilde{x}_k}{m} \right\rfloor \right)$$



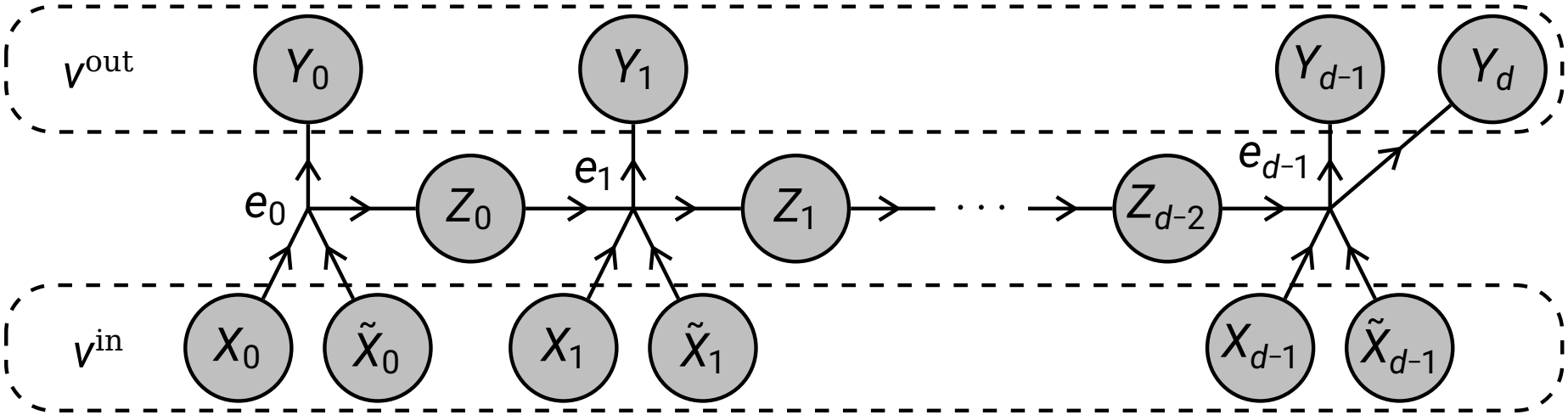
Decomposition of the basis encoding

We have a decomposition of the basis encoding of the sum as

$$
\begin{aligned}
&\beta^{+^m}\left[Y_{[d+1]}, X_{[d]}, \tilde{X}_{[d]}\right] \\
=\big\langle &\{\beta^{\tilde{+},0}\left[Y_0, Z_0, X_0, \tilde{X}_0\right]\} \cup \\
&\{\beta^{\tilde{+},k}\left[Y_k, Z_k, X_k, \tilde{X}_k, Z_{k-1}\right] \,:\, k \in \{1, \ldots, d-2\}\} \cup \\
&\{\beta^{\tilde{+},d-2}\left[Y_{d-1}, Y_d, X_{d-1}, \tilde{X}_{d-1}, Z_{d-2}\right]\}\big\rangle_{\left[Y_{[d+1]}, X_{[d]}, \tilde{X}_{[d]}\right]}.
\end{aligned}
$$

Tensor Network representation

a) Hypergraph representing the circuit:



b) Basis encoding with decomposition: